

It from Anomalies

Universal Defect Energy and Entropy from Weyl Anomalies

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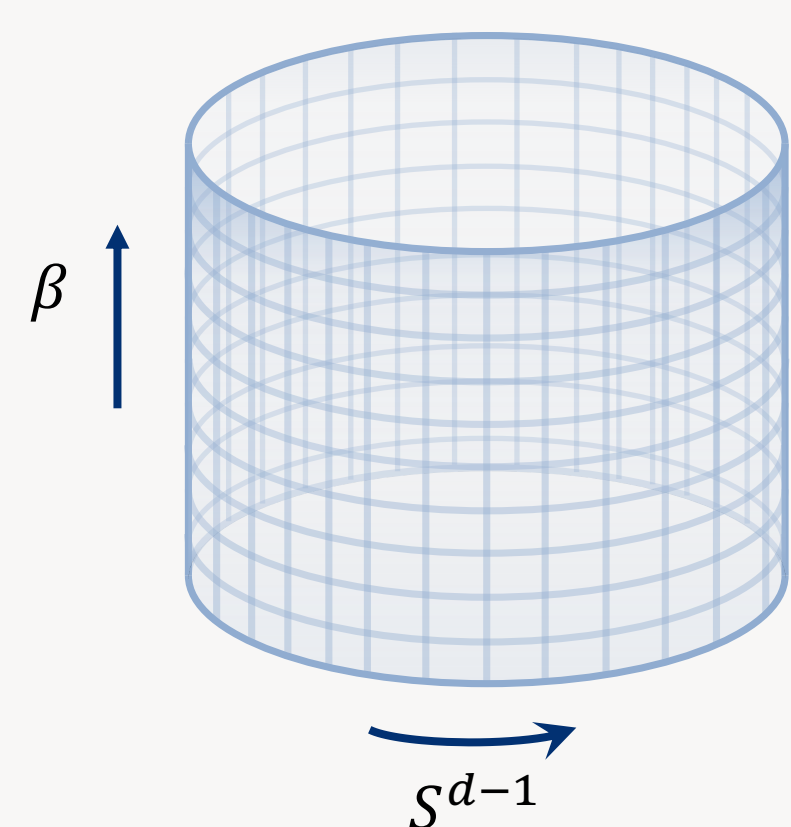
MOTIVATION

- **Weyl anomaly** (central charges) governs the entanglement structure and ground state energy of CFTs

$$2D \text{ CFT} \quad \langle T^\mu_\mu \rangle = -\frac{c}{24\pi} R \quad E = -\frac{c}{12} \quad S_n = \frac{c}{6} \left(\frac{1}{n} + 1 \right) \log \frac{l}{\epsilon}$$

- Defects have **non-trivial interactions** with ambient theory

$$\langle t^\mu_\mu \rangle = -\frac{1}{24\pi} \left[b R^\Sigma + d_1 \Pi^\mu_{ab} \Pi^\mu_{ab} - d_2 W_{ab}^{ab} \right]$$



Defect Anomalies

$$b, d_2, k_R, \dots$$

(Weyl/ 't Hooft)

Universality

Structural Relation

Defect Energy

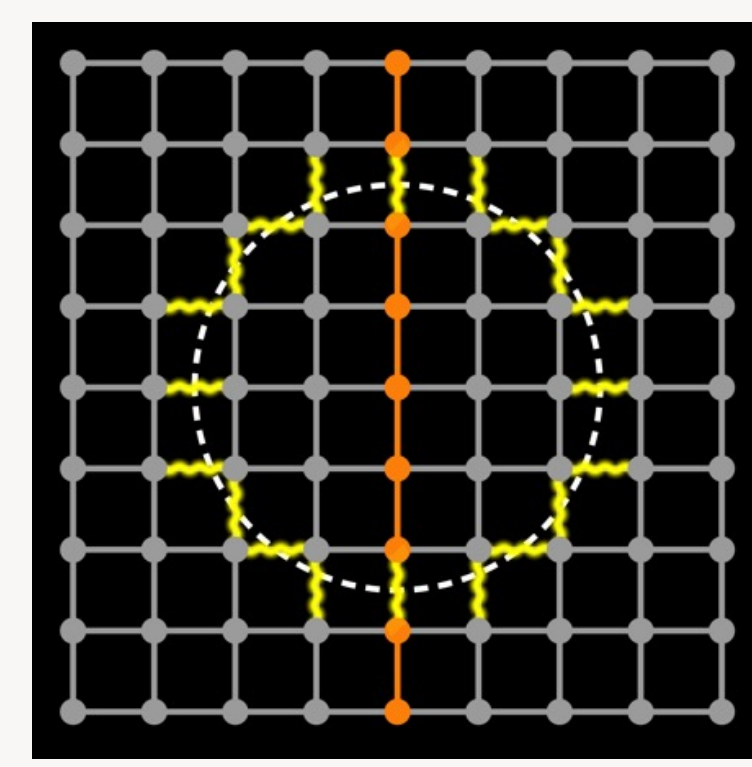
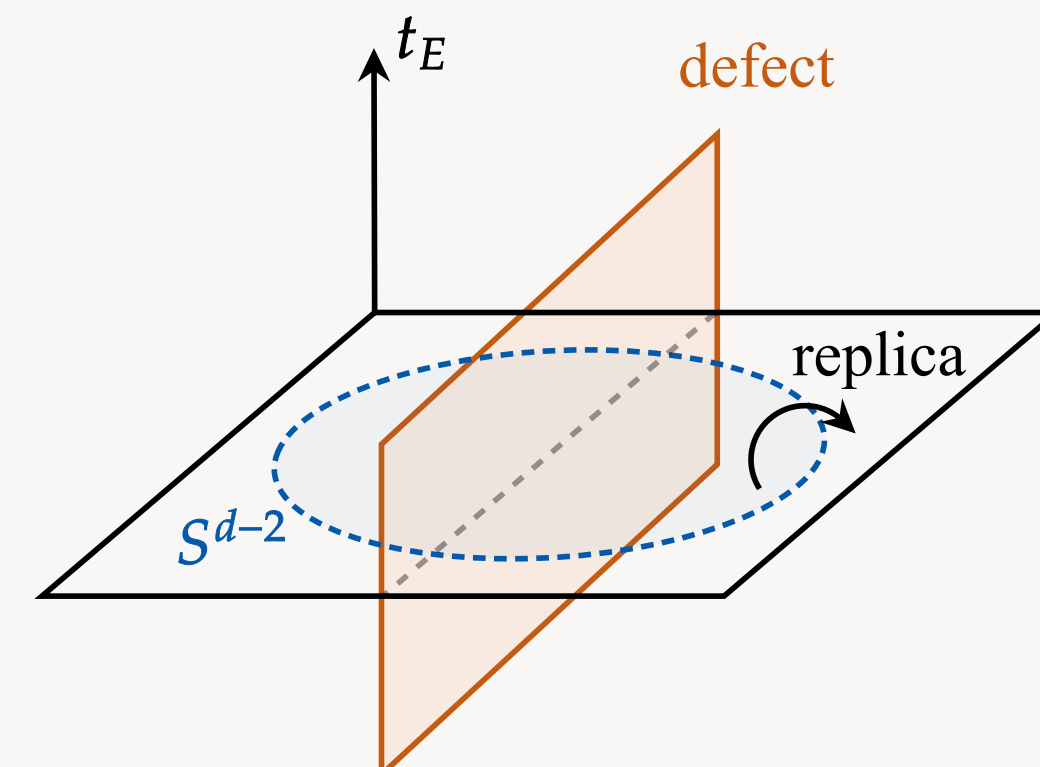
Defect contribution to the twisted Casimir energy (dSCE)
 $\log \langle D \rangle = -\beta E_{\mathcal{R}} + O(\beta^0), \quad \beta \rightarrow \infty$

MAIN QUESTION

- ❓ **With defect**, is there any analogous **universal relation** between **anomaly, energy & entropy**?

MAIN RESULT

Surface defect contribution to the SUSY Casimir energy (**dSCE**) and the SUSY Rényi entropy (**dSRE**) are **controlled by defect anomaly data**.



Defect Entropy

Defect contribution to the twisted Rényi entropy (dSRE)

$$S_n = \frac{1}{1-n} \log \frac{Z_n}{(Z_1)^n}$$

CLOSED FORMULAE

Illustrative example: 1/2-BPS Wilson surface in 6d (2,0) SCFTs [1]

$$E_{\mathcal{R}} = -\frac{1}{\omega_1} \left[\frac{d_2 - b}{6} \omega_2 \omega_3 + \frac{2b - d_2}{24} \sigma_1 \sigma_2 \right]$$

$$S_n = \frac{2b - d_2}{6} \left[\frac{r_1 r_2}{2} \left(\frac{1}{n} - 1 \right) + 1 \right] \log \frac{l}{\epsilon}$$

Partition function in radial quantization

$$Z = \text{Tr} \left[e^{-\beta \sum_{i=1}^3 \omega_i I_i + \sum_{a=1}^2 \sigma_a R_a} \right]$$

$\omega_{1,2,3}$	angular momentum fugacity
$\sigma_{1,2}$	R-symmetry fugacity
$r_{1,2}$	weights of R-symmetry fugacity

THREE-IN-ONE METHOD

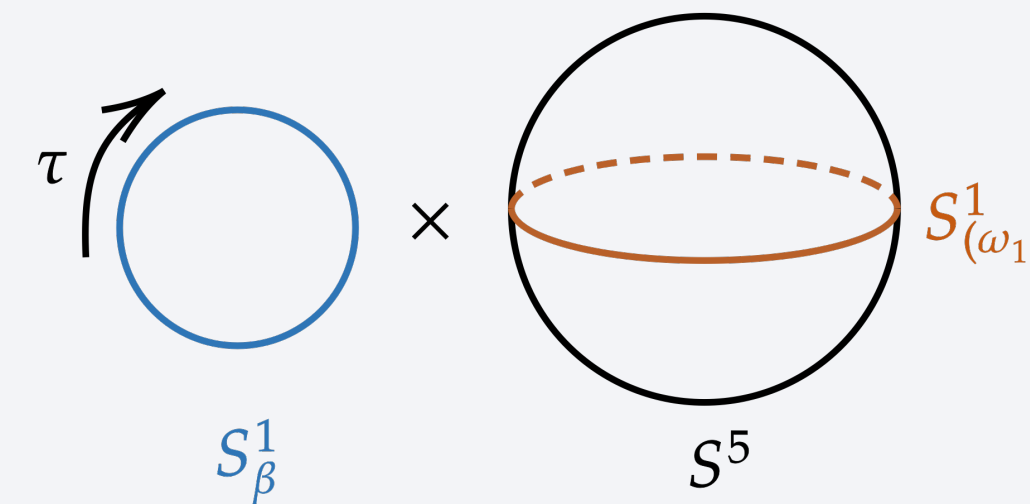
1 Localization

The large N behavior of dSCE

The $n \rightarrow 0$ behavior of dSRE

Dimensional reduction $\Rightarrow$ 5d MSYM matrix integral with Wilson loop

$$\langle D \rangle = \frac{1}{Z} \int dv_i \exp \left[\frac{2\pi^2}{\omega_1 \omega_2 \omega_3} \left(-\frac{1}{\beta} \sum_{i=1}^N v_i^2 - \frac{\sigma_1 \sigma_2}{2} \sum_{j<i} (v_i - v_j) \right) \right] \text{Tr}_{\mathcal{R}} \exp \left[\frac{2\pi v}{\omega_1} \right]$$



Saddle point approximation at large β

$$E_{(k)} = -\frac{\beta}{2\omega_1} [k^2 \omega_2 \omega_3 + k(N-1) \sigma_1 \sigma_2] \quad E_{[k]} = -\frac{\beta}{2\omega_1} [k \omega_2 \omega_3 + k(N-k) \sigma_1 \sigma_2]$$

Geometry relates the dSCE & dSRE by energy density $S_{n \rightarrow 0} = -\frac{\text{Vol}[\mathbb{H}^1]}{\text{Vol}[S^1_\beta]} \beta E_{\mathcal{R}} \Big|_{n \rightarrow 0} = \frac{2b - d_2}{12} \frac{r_1 r_2}{n} \log \frac{l}{\epsilon}$

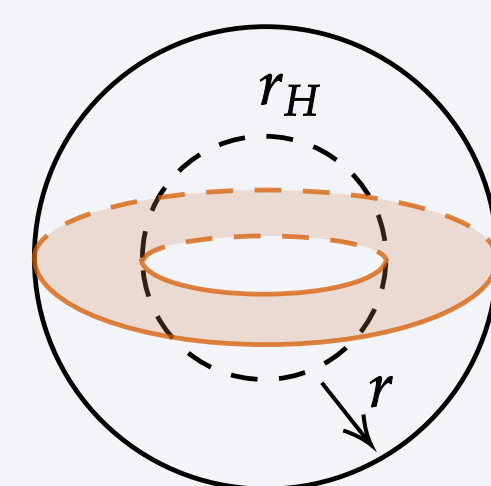
2 Holography

The large N behavior of dSRE

Defect in fundamental representation $\xleftrightarrow{\text{AdS}_7/\text{CFT}_6}$ Probe M2-brane in 7d two-charge topological black hole

$$\log \langle D \rangle = -I_{M2} = \pi n T_2 r_H^2 \text{Vol}[\mathbb{H}^1]$$

$$S_n(N \rightarrow \infty) = \frac{\log \langle D \rangle - n \log \langle D \rangle|_{n=1}}{1-n} = 2N \left[\frac{r_1 r_2}{2} \left(\frac{1}{n} - 1 \right) + 1 \right] \log \frac{l}{\epsilon}$$



3 Anomaly Polynomial

Exact dSCE and dSRE

Consistency check

Generalize the SCFT method to the defect insertion

$$A_4 = \frac{1}{2} (\lambda, \lambda) (c_2(L) - c_2(R)) + (\lambda, \rho) (c_2(I) - c_2(F))$$

$$\left\{ \begin{array}{l} \text{Equivariant integral} \quad E = - \int_{\mathbb{R}^2_{\omega_1}} A_4 \\ \text{Extended integral} \quad \tilde{E} = - \int_{\mathbb{R}^2_{\omega_1} \times \mathbb{R}^2_{\omega_4=1/n}} A_4, \quad S_n = \frac{n \tilde{E}|_{n=1} - \tilde{E}|_n}{1-n} \text{Vol}[\mathbb{H}^1] \end{array} \right.$$

More defect systems [2]

- Gukov-Witten defect in 4d N=4 SYM

$$E = -k_R \frac{\Delta_2 \Delta_3}{\omega_1}$$

$$S_n = k_R \left[s_1 s_2 \left(\frac{1}{n} - 1 \right) + 1 \right]$$

- Surface defect in 6d (1,0) SCFTs

$$E = k_I \frac{a_I^2}{8\omega_1} + k_g \frac{\omega_1}{24}$$

$$S_n = \frac{1}{24} \left(\frac{k_g + 3k_I}{n} + (k_g + 21k_I) \right)$$

Reference

[1] Z.-X. Huang, M.-K. Yuan and Y. Zhou, *From Weyl Anomaly to Universal Defect Casimir Energy and Rényi Entropy*, **Phys. Rev. Lett.** **136** (2026) 201601.

[2] Z.-X. Huang, M.-K. Yuan and Y. Zhou, *Defect Super-symmetric Casimir Energy and Rényi Entropy in 4d and 6d* (to appear).

